SCD - Slowly Changing Dimension

SCD (Slowly Changing Dimension) is used to manage changes in Dimension tables over time.

SCD Types:
1. SCD Type 1 – Overwrite the old data. No history is maintained.
2. SCD Type 2 – Insert a new record for every change. Full history is maintained using:
- Surrogate Key
- Effective Start Date
- Effective End Date
- Is_Active flag
3. SCD Type 3 – Maintain limited history using additional columns (e.g., previous value).

Fact vs Dimension:

Fact Table:
- Stores measures, quantities, and metrics
- Contains foreign keys
- Mostly insert-only
- Represents business events

Dimension Table:
- Stores descriptive attributes (Who, What, Where, When)
- Changes over time
- SCD is applied here

Banking Domain Example:

Dimension Tables:
- Customer
- Account

Fact Table:
- Transaction

SCD Type 2 Handling:

Update Scenario:
When a customer changes mobile number:
- Old record is marked inactive
- End_Date is updated
- New record inserted with new Start_Date

Insert Scenario:
New customer onboarding → Insert new record with Start_Date and Is_Active = Y

Delete Scenario:
No physical delete.
Instead:
- Is_Active = N
- End_Date updated
- Optional Is_Deleted flag

Point-in-Time Reporting Example:

To find active customers on Day 20:

```
SELECT COUNT(*)
FROM dim_customer
WHERE 'Day20' BETWEEN start_date AND end_date;
```

Data Modeling Levels:

1. Conceptual Data Model – High-level entities (Customer, Account, Transaction)
2. Logical Data Model – Attributes, Primary Keys, Relationships (1:1, 1:M, M:M)
3. Physical Data Model – Actual tables, data types, indexes, surrogate keys, SCD columns